

Nuclear#1 // Homework#1 Parker Lewis

1.) harmonic oscillator:

$$n \geq 1; \quad E_n = (n - 1 + \frac{1}{2}) \hbar \omega \quad \begin{matrix} * \text{from} * \\ \text{notes} \end{matrix}$$

$$l = n - 1, n - 3, \dots \geq 0$$

$$j = l \pm \frac{1}{2}, \quad |l - s| \leq j \leq |l + s|$$

$$n = 7$$

$$l = 1, \dots$$

$$n = 7:$$

$$l = 1, 3, 5, 7$$

$l = 1:$	$l = 3$	$l = 5$
$j = \frac{1}{2}, \frac{3}{2}$	$j = \frac{5}{2}, \frac{7}{2}$	$j = \frac{9}{2}, \frac{11}{2}$

$$l = 7$$

$$j = \frac{13}{2}, \frac{15}{2}$$

$$j = \frac{14 - 1}{2}, \frac{14 + 1}{2}$$

Use d:

$$j = l \pm \frac{1}{2}$$

and

$$N = 8$$

$$n = 0, 1, 2, 3, 4, 5, 6, 7$$

$$l = n - 1, n - 3, \dots \geq 0$$

Hw #2

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(week 2 "Bonner/Paylich exp")
2.) • Our previous paper - the bonner experiment
1.) proves magic number for $A=3$
Nuclei because ${}^3\text{He}$ and ${}^3\text{H}$
they don't have an excited state (not determined by deexcited gammas)
from bonner experiment one of the reasons
is that at least for $Z=2$ or $N=2$
Combinations of ${}^3\text{He}$ and/or ${}^3\text{H}$ that
is a magic number filling which
makes these isotopes very stable in
ground state.

2.) Magic number in light nuclei
are stable, there may be
magic numbers within heavy nuclei
that are more stable than nearby
neighbors have very long-lifetimes
this is all proposed would create
"island of stability".

Citation: <https://arxiv.org/pdf/2003.12720>

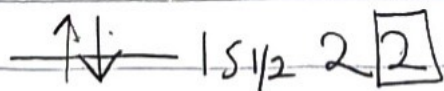
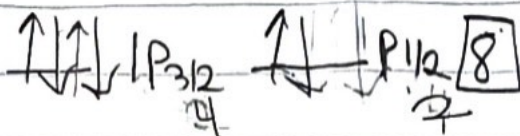
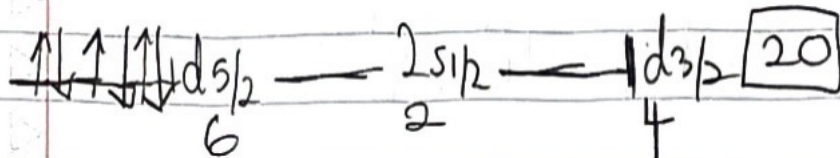
Hw # 2

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3.)

^{29}Si

$$Z = 14, N = 15$$



Π

$$\Pi (1s_{1/2})^2 (1p_{3/2})^4 (1p_{1/2})^2 (1d_{5/2})^6$$

$$N = 15$$

$$\begin{array}{c} \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \\ 6 \end{array} d_{5/2} \quad \begin{array}{c} \uparrow \\ 2 \end{array} 2s_{1/2} \quad \begin{array}{c} \text{---} \\ 4 \end{array} 1d_{3/2} \quad \boxed{20}$$

$$\begin{array}{c} \uparrow\downarrow \uparrow\downarrow \\ 4 \end{array} 1p_{3/2} \quad \begin{array}{c} \uparrow\downarrow \\ 2 \end{array} 1p_{1/2} \quad \boxed{8}$$

$$\begin{array}{c} \uparrow\downarrow \\ 2 \end{array} 1s_{1/2} \quad \boxed{2}$$

V

$$V \left(1s_{1/2} \right)^2 \left(1p_{3/2} \right)^4 \left(1p_{1/2} \right)^2 \left(1d_{5/2} \right)^6 \left(2s_{1/2} \right)^1$$

$$s(l=0) \quad p(l=1) \quad d(l=2) \\ f(l=3)$$

Ground state:

$2s_{1/2}$

$$J = 1/2, \quad \pi = (-1)^l$$

~~$J^\pi = 1/2^+$ $l=0$~~
from nndc $\rightarrow J^\pi = 1/2^+$ ground state

1st-excited state:

$$J = 3/2 \quad 1d_{3/2} \quad l=2 \quad \pi = (-1)^2 = 1$$

$J^\pi = 3/2^+$ from nndc $J^\pi = 3/2^+$

2nd - excited State:

$$1 d_{5/2}$$

$$J = 5/2 \quad l = 2$$

$$\pi = (-1)^2 = 1 = +$$

$$J^\pi = 5/2^+$$

from nndc $J^\pi = 5/2^+$

1) Reasoning for ground state is simply that the unpaired neutron at $5/2$ makes it the ground state and why $J^\pi = 1/2^+$

2) Reason for first excited is that $d_{3/2}$ is right within sd shell so it's favorable for $J^\pi = 3/2^+$ still $d \Rightarrow l=2$

3) 2nd excited is when $1 d_{5/2}$ because $f_{7/2}$ has larger bandgap so more favorable to have neutron to

next page 

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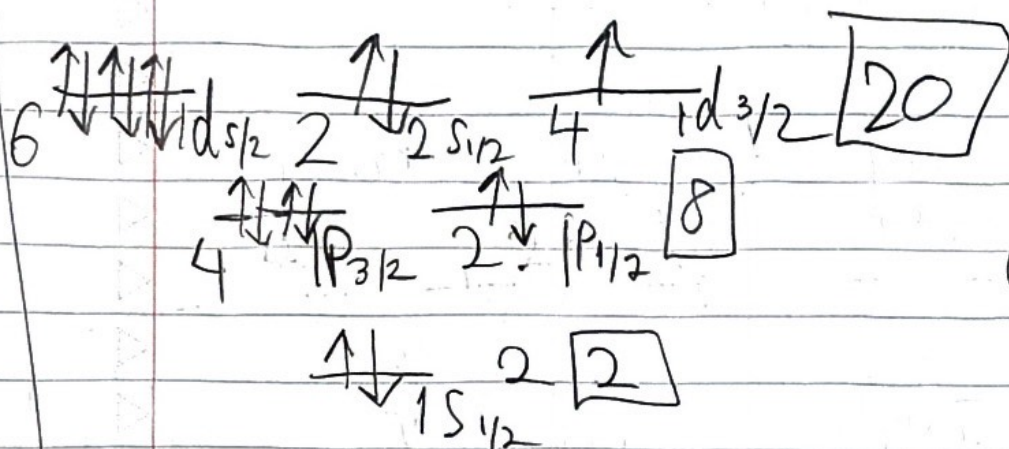
from $1d_{5/2}$ to $1d_{3/2}$ or $2s_{1/2}$ Both combinations of promotions could be feasible because of non-large gap and in same sd-subshell it may be more feasible for $1d_{3/2}$ because their feasibility of pairing spins/orbital momenta.

4.)

^{34}Cl

$$Z = 17$$

$$N = 17$$



(Proton)

$$\pi (1s_{1/2})^2 (1p_{3/2})^4 (1p_{1/2})^2 (1d_{5/2})^6 (2s_{1/2})^2 (1d_{3/2})^1$$

$\uparrow\uparrow\uparrow\uparrow$ 6 $1d_{5/2}$ $\frac{1}{2}$ $\uparrow\downarrow$ 2 $2s_{1/2}$ $\frac{1}{4}$ $\uparrow$ $1d_{3/2}$ $\boxed{20}$ (neutron) 5

$\uparrow\uparrow\uparrow\downarrow$ 4 $1p_{3/2}$ $\frac{1}{2}$ $\uparrow\downarrow$ 2 $1p_{1/2}$ $\boxed{8}$

$\uparrow\downarrow$ 1 $5s_{1/2}$ 2 $\boxed{2}$

$$\sqrt{
 \begin{pmatrix} 1 \ 5_{1/2} \end{pmatrix}^2
 \begin{pmatrix} 1 \ p_{3/2} \end{pmatrix}^4
 \begin{pmatrix} 1 \ p_{1/2} \end{pmatrix}^2
 \begin{pmatrix} 1 \ d_{5/2} \end{pmatrix}^2
 \begin{pmatrix} 2 \ s_{1/2} \end{pmatrix}^2
 }
 \begin{pmatrix} 1 \ d_{3/2} \end{pmatrix}^1$$

JJ-coupling Rule for $\pi \frac{1}{2} V$

$$J = 3/2 = 2 - 1/2 \rightarrow l_1 - s_1$$

$$J = 3/2 = 2 - 1/2 \rightarrow l_2 - s_2$$

$$\text{Rule \#2 } J = |J_1 \pm J_2|$$

$$J = \frac{3}{2} - \frac{3}{2} = 0$$

$$\pi = (-1)^2 (-1)^2$$

(Parity)

$$l = 2$$

Ground State: $J^\pi = 0^+$

from nuc $J^\pi = 0^+$, reason: using JJ-rule #2 it works.

1st

- Test Neutron $1d_{3/2} \rightarrow 1f_{7/2}$
* Rule #1 must be applied because

$$\pi, \frac{3}{2} = J = 2 - \frac{1}{2} \rightarrow J_1 = L_1 - S_1$$

$$V, \quad J = \frac{7}{2} = 3 + \frac{1}{2}, \quad J_2 = L_2 + S_2$$

$$J_1 = \left| \frac{3}{2} - \frac{7}{2} \right| = \left| \frac{-4}{2} \right| = 2$$

$$\pi = (-1)^{L_1} (-1)^{L_2} = -$$

$$J^\pi = 2^-$$

neutron $2s_{1/2} \rightarrow 1d_{3/2}$

$$\begin{aligned} \pi &\rightarrow J = 3/2, J = 7/2, J_1 = l_1 - S_1 \\ \nu &\rightarrow J = 3/2, J = 2 - 1/2, J_2 = l_2 - S_1 \end{aligned}$$

Rule # 2 $J = 0$

$$\pi = +$$

~~* (earlier example) *~~

* Using JJ rules with possibilities that work for 1st excited state works both for proton and neutron then

$$1d_{3/2} \rightarrow 1f_{7/2} \quad J^\pi = 2^-$$

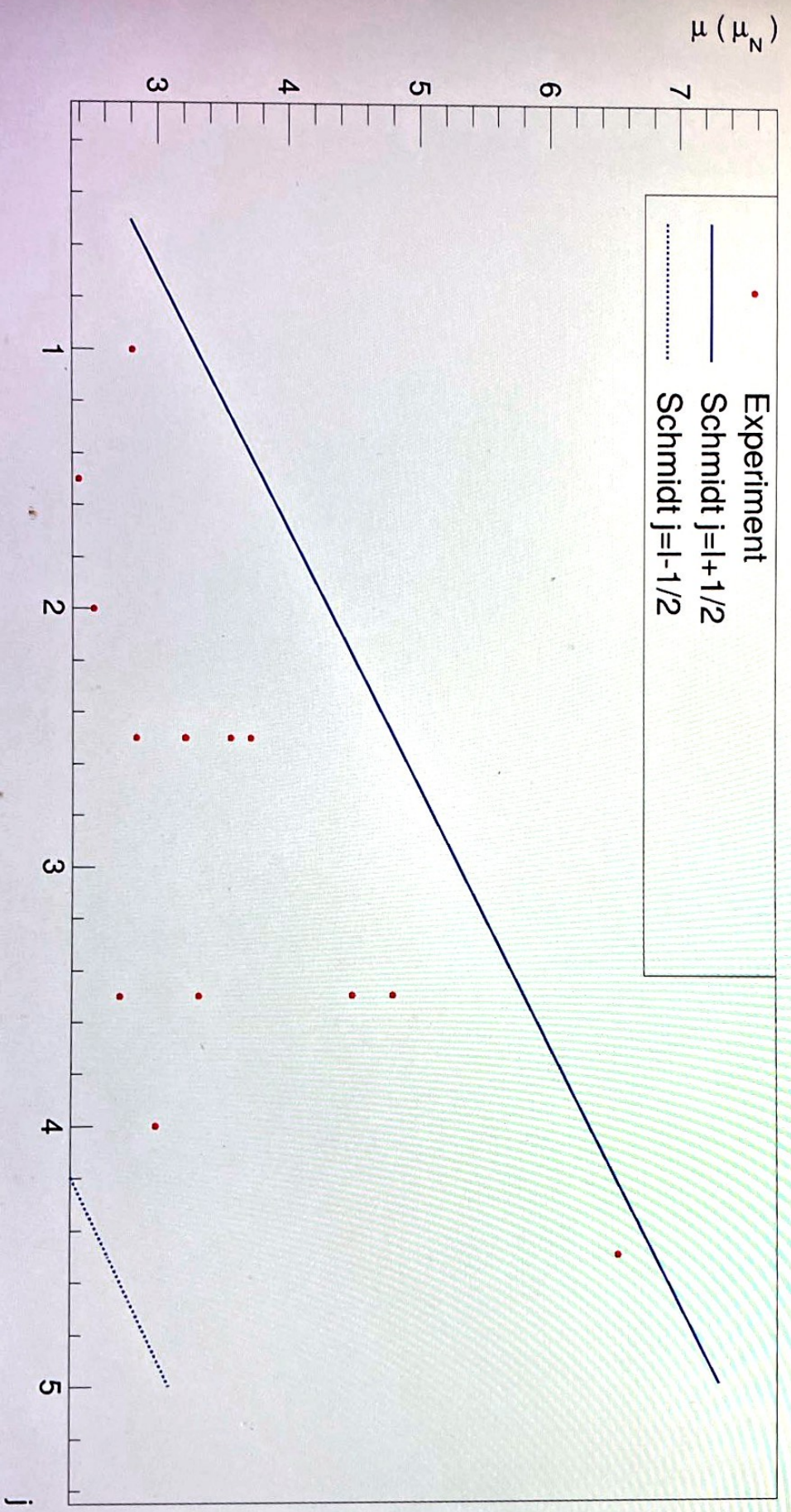
$$2d_{5/2} \rightarrow 1d_{3/2} \quad J^\pi = 0^+$$

from nndc $J^\pi = 3^+$ which means it's shows limitation of shell model / predicting spin and parities

Also for 2nd excited state nndc $J^\pi = 1^+$ which then could be

multitude of raising for 2nd excited state that then shows also inconsistency for shell model to predict

Odd-Z μ vs Schmidt limits



AI Usage: Pseudo code, how to input
and the file into arrays,
parsing.

5.) Resulting thoughts:

The magnetic moment either
overshoot or undershoot
each Schmidt limit line.
Indicating more complex
structures than
the single-particle mean-field
assumptions used to build
shell model.

6.) ideal axial rotor:

$$E_{rot} = \frac{J(J+1)\hbar^2}{2I}$$

• $I \rightarrow$ moment of inertia

• Substituted constant: $A = \frac{\hbar^2}{2I}$

$$E(J) = AJ(J+1)$$

$$E(4^+) = 20A$$

$$E(2^+) = 6A$$

$$\text{Ratio} = \frac{E(4^+)}{E(2^+)} = \frac{10}{3} \approx 3.33$$

- Ideal Ratio ≈ 3.333
("axial ratio")

- ^{226}Ra
 $E(2^+) = 68 \text{ keV}$
 $E(4^+) = 211 \text{ keV}$

ExR Ratio $\frac{E(4^+)}{E(2^+)} \approx \frac{211}{68} \approx 3.103$

Ratios closely approximately match
there we can approx say we have Ryd Ratio

$$E(6^+) = 42 \text{ A} \quad E(2^+) = 6 \text{ A}$$

$$\frac{E(6^+)}{E(2^+)} = 7$$

$$E(6^+) = E(2^+) \times 7 = 68 \times 7 = 476 \text{ keV}$$

$$\text{From nndc } E(^6\text{Li}) = 46.5 \text{ keV}$$

$$^{32}\text{Mg}, \beta = 0.51$$

$$A=32$$

$$E(J) = J(J+1)\hbar^2/2I$$

Spherical moment of inertia in book:

$$I = \frac{2}{5} MR^2 (1 + 0.31\beta)$$

$$.) m = A \cdot u$$

$$.) R = r_0 A^{1/3} = 1.2 \times 10^{-15} \times A^{1/3} = 1.2 \times 10^{-15} \times (32)^{1/3}$$

$$.) R = 3.81 \times 10^{-15} \text{ m}$$

$$.) m = 32 u \left(\frac{1.66 \times 10^{-27} \text{ kg}}{u} \right) = 5.312 \times 10^{-26} \text{ kg}$$

$$.) \hbar = 1.054 \times 10^{-34} \text{ J}\cdot\text{s}$$

$$.) I = 3.572 \times 10^{-55} \text{ kg}\cdot\text{m}^2$$

$$.) \hbar^2/2I = \frac{2 (1.054 \times 10^{-34})^2}{2 (3.572 \times 10^{-55})} \frac{\text{J}^2\cdot\text{s}^2}{\text{kg}\cdot\text{m}^2} = 1.56 \times 10^{-14} \text{ J}$$

$$\frac{h^2}{2I} = 1.56 \times 10^{-14} \text{ J}$$

$$1) \quad 1.602 \times 10^{-13} \text{ J} = 1 \text{ MeV}$$

$$1) \quad \frac{h^2}{2I} = 1.56 \times 10^{-14} \text{ J} \quad \frac{1 \text{ MeV}}{1.602 \times 10^{-13} \text{ J}}$$

$$\frac{h^2}{2I} = 0.097 \text{ MeV}$$

$$E(2^+) = 2(3)(0.097 \text{ MeV}) = 0.582 \text{ MeV}$$

$$E(4^+) = 4(5)(0.097 \text{ MeV}) = 1.94 \text{ MeV}$$

$$E(6^+) = 6(7)(0.097 \text{ MeV}) = 4.074 \text{ MeV}$$

Comparison to paper:

$$E(2^+) = 0.886 \text{ MeV}$$

$$E(4^+) = 2.324 \text{ MeV}$$

$$E(8^+) = 4.097 \text{ MeV}$$

8) ^{173}Lu $\beta = 0.306$

$Z=71$ $N=102$

Shell model:

$$\frac{\pi(1h_{11/2})^{11}}{\nu(3p_{3/2})^{24}}$$

$$J_1 = 11/2$$

$$J_2 = 3/2$$

$$l_p = 1$$

$$l_n = 5$$

* Proton are only one, left *

$J^\pi = 11/2^-$ (shell model) unpaired

$$\pi(\text{parity}) = (-1)^5 = -$$

Nelson model: $\beta = 0.306$

If protons are moved up 21 pro
2-states then will stop $g_{7/2}$
at $J^\pi = 7/2^+$, mdc $J^\pi = 7/2^+$

9.) - Proton for this case
Fermi Gas

Using relativity:

$$\bullet \beta = \frac{v}{c} = \frac{pc}{E} \quad (\text{boost relations})$$

$$\bullet \frac{E^2}{c^2} - p^2 = m^2 c^2$$

$$\bullet E^2 = p^2 c^2 + m^2 c^4$$

$$\bullet p_F = \hbar k_F \quad \text{de-broglie relation}$$

$$\bullet k_F = 1.339 \text{ fm}^{-1}$$

$$p_F =$$

10.) Ungrading I would put myself
for this assignment
just because for 75-82 range
to be able to put into problem.
for this Hw and their may be
math mistakes because I was
going fast, this section was a
more challenging.